

Abstract

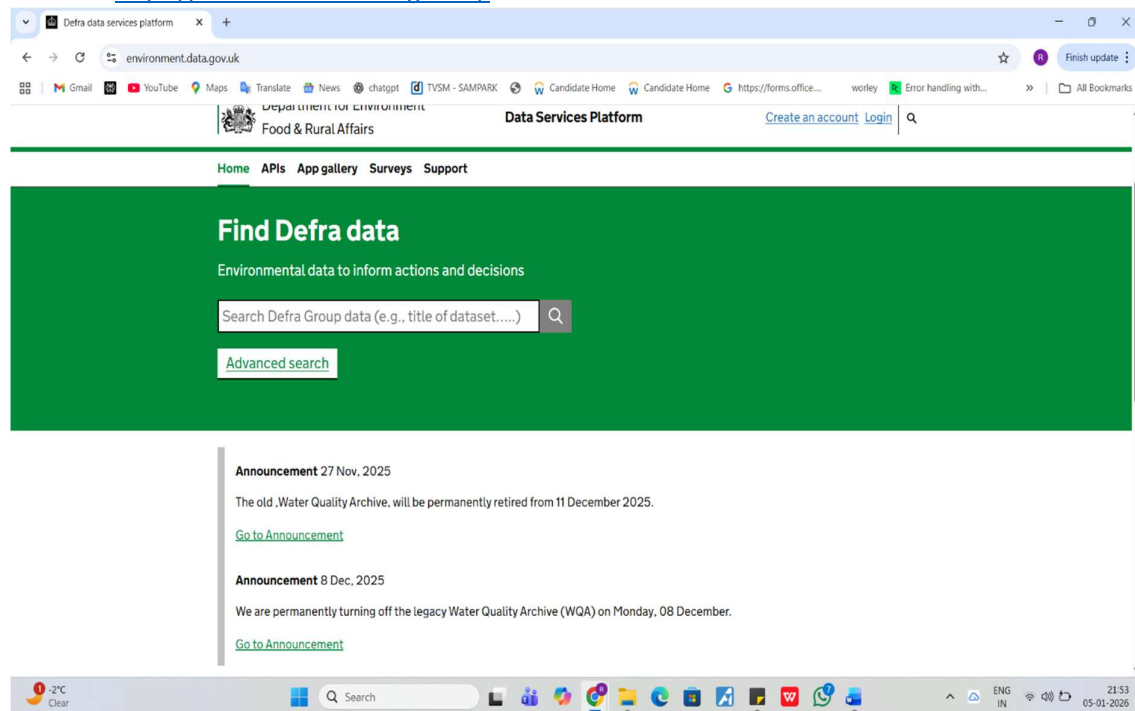
Water pollution monitoring statistics would be utilized on a retroactive basis and in this case, have limited utility in proactive management of the environment. The project creates a decision-support framework that can be interpreted and integrates a time-series forecasting framework and pollution risk assessment framework.

Based on the river water quality data of 2020 to 2025, the key determinands are subsequently scaled into a composite Pollution Pressure Index (PPI). SARIMA and LSTM predict the values of future PPI and a rule based layer converts quantitative risk level predictions into qualitative risk levels. The findings show that regular monitoring information could be converted into a useful early-warning system to assist in environmental decision-making.

Source for Data: <https://www.kaggle.com/datasets/ranjinik31/environmental-agency-water-quality/data>

The Official Link seems to be facing technical issues:

Old Link: <https://environment.data.gov.uk/>



New Link:

<https://www.data.gov.uk/search?q=water+quality+archive&filters%5Bpublisher%5D=Environment+Agency&filters%5Btopic%5D=Environment&filters%5Bformat%5D=&sort=best>

